

Chapter 12 — Fundamentals of Intraoperative Patient Safety

- **"First do no harm"**: treating patients carries the duty to minimise risk.
- **Surgery is a team effort** inside hospital systems larger than any individual, yet **the surgeon remains team leader and ultimately responsible** when complications arise.

BACKGROUND

- **Institute of Medicine (IOM), *To Err Is Human* (1999)**, landmark: **44,000–98,000 deaths and >1 million injuries a year** from medical error → catalysed patient-safety research.
- **A 2016 report** judged this a severe underestimate: **medical error as the third leading cause of US death**, after heart disease and cancer.
- **~50% of health care adverse events involve surgery, and ~50% of those are preventable.**
- **Closed malpractice claims**: three-quarters of errors arose wholly or partly in the **intraoperative phase** — making the operating room (OR) error-free is indispensable.

SURGICAL TIME-OUTS AND SAFETY CHECKLISTS

- **Complexity plus stress and fatigue make human error almost inevitable.**
- **A checklist** lists tasks, components or criteria required for a task; it is a **mnemonic device** that aids recall, standardises steps and ensures best-practice adherence — borrowed from aviation.
- **Pronovost (2006)**, landmark: a central-line checklist across **108 intensive care units** cut catheter-related bloodstream infection by **66%** over 18 months.
- **World Health Organization (WHO) Surgical Safety Checklist (2008)**, **19 items** (Fig. 12.1); pilot at 8 institutions worldwide (~4000 operations): **death ↓ 47%, major complications ↓ 36%**.
- **Institutions add** antibiotic and venous thromboembolism (VTE) prophylaxis, normothermia and other items; **some form of checklist is now standard everywhere.**

FIG. 12.1 WORLD HEALTH ORGANIZATION SURGICAL SAFETY CHECKLIST (FIRST EDITION)

Sign in — before induction	Time out — before skin incision	Sign out — before patient leaves theatre
Patient confirms identity, site, procedure, consent; site marked/not applicable; anaesthesia safety check; pulse oximeter on and working; known allergy?; difficult airway/aspiration risk (equipment ready)?; risk of >500 mL blood loss (7 mL/kg in children) — access and fluids planned?	All team introduce themselves by name and role; surgeon, anaesthetist and nurse confirm patient, site, procedure; anticipated critical events — surgeon (critical steps, duration, blood loss), anaesthesia (patient concerns), nursing (sterility, equipment); antibiotic prophylaxis within the last 60 minutes? ; essential imaging displayed?	Nurse verbally confirms: procedure name recorded; instrument, sponge and needle counts correct ; specimen labelled (including patient name); equipment problems to address; team reviews key concerns for recovery

- **Pattern**: three pauses tied to three transitions — induction, incision, exit; the checklist is intended to be adapted locally, not treated as comprehensive.

WRONG-SITE SURGERY PREVENTION

- **Safety starts with correct patient, procedure and site.** Wrong-site surgery = **wrong side, wrong patient or wrong part** — all **sentinel, reportable never events**.
- **Extremely rare (~1 in 112,000 operations)** but devastating.
- **The Joint Commission** (independent not-for-profit, founded 1951, accredits >20,000 US organisations) published the mandatory **Universal Protocol (2004)**, by expert consensus, with **three components**:
preprocedure verification → surgical site marking → time-out immediately before starting
It is now the foundation of intraoperative checklists.

Preprocedure Verification Process

- **Up to 30% of wrong-site root causes arise before the patient reaches hospital:** wrong or incomplete clinic notes, mislabelled imaging, laboratory mix-ups of similar names.
- **Confirms three things:** correct identity, correct procedure, exact site.
- **Review** notes, allergies, imaging and laboratory results for correct, relevant indication.
- **Identify with multiple identifiers** — full name, date of birth, medical record number — on a durable tracking system (barcoded wristband).
- **History and examination recent** and updated for interim change.
- **Consent form** names the responsible surgeon and procedure, signed by patient or designate (reconfirmed if signed earlier); **list contingent procedures** that depend on intraoperative findings.
- **Use professional medical interpreters** — **never family** — for procedural consent.
- **Resolve any missing or discrepant information before proceeding.**

Site Marking

- **Required where more than one site is possible** (e.g. unilateral radical nephrectomy, radical orchidectomy).
- **Mark must be clear, unambiguous, permanent enough to survive skin preparation;** by the responsible surgeon or delegated resident/advanced practice provider — ideally someone **present at the time-out**.
- **Mark with an awake, alert patient** before sedation where possible.
- **Mark only the correct site** — **never "No" or "X" on the wrong side**.
- **Lower-tract endourology:** mark sidedness somewhere visible (e.g. wristband), not the urethral meatus.
- **Barriers:** delegation to someone absent from the case; marking the wrong site after erroneous record review; marking dressings, not skin; non-permanent markers; persistent old marks (take-backs); site impossible to mark → **hospital-approved alternative process**.

Time-Out

- **A purposeful brief pause just before incision or the invasive step,** confirming identity, site and procedure.
- **Extended versions:** normothermia, euglycaemia, β -blockade, antibiotics, spinal precautions, critical steps, duration, blood need, fire risk, VTE prevention, and "any concerns?".
- **All members participate actively;** leader (surgeon or circulating nurse) is institution-specific; no interruptions or side conversations; proceed only with full agreement.
- **Repeat the time-out** at every **change of surgeon** or transition to a **separate procedure** (multispecialty cases).

- **Common errors:** background noise (music) continues; proceeding while a member is mid-task; key member on break; going through the motions ("we all know the patient").

Concept — *The Universal Protocol is layered redundancy: verification catches errors made before the day, marking fixes the side on the body, and the time-out forces a shared final check. Each layer is only as good as the attention paid to it, which is why "going through the motions" is the characteristic failure.*

SYSTEMS FAILURES

- **"Swiss cheese model"** (Fig. 12.2): each barrier is a slice with unpredictable holes; hazards passing one hole are usually stopped by the next (**near miss**); **harm occurs when the holes align**.
- **Layers (Fig. 12.2):** organisational influences → unsafe supervision → preconditions for unsafe acts (**latent failures**) → unsafe acts (**active failures**) → error.
- **Explains why never events persist** despite the Universal Protocol: many stakeholders, departments and levels — harm is usually systemic.
- **Risk circumstances for wrong-site surgery:** emergencies; unusual anatomy (deformity, morbid obesity); time pressure; multiple surgeons; multiple procedures in one sitting.
- **Rushed situations demand the checklist most;** checklists do not delay start times, and in urgent operations they **significantly reduce complications and mortality**.
- **Environmental root causes:** hierarchy and power dynamics that silence concern; staffing shortages and temporary staff untrained in local protocols; unprecedented events (SARS-CoV-2) forcing real-time workflow change.

Concept — *Errors reaching the patient are system failures, not individual lapses: no single barrier is reliable, so safety comes from several independent barriers and from preventing the latent conditions — hierarchy, staffing, time pressure — that make the holes line up.*

RETAINED SURGICAL ITEMS

- **Retained surgical item (RSI):** material, instrument or equipment unintentionally left in the patient; preferred to "retained foreign object" (distinguishes from shrapnel or bullets).
- **Consequences:** infection, pain, death; medicolegal liability; cost.
- **Joint Commission definition:** an RSI if discovered after skin closure — even with the patient still anaesthetised in theatre.
- **A never event,** theoretically 100% preventable, yet occurring **between 1 in 10,000 and 1 in 2800 operations**.

Rank	Category	Commonest item
1st	Surgical sponges (~two-thirds of RSIs)	Laparotomy pads and small swabs — small, conformable, invisible when blood-soaked
2nd	Drains and catheters	Penrose drain
3rd	Needles and blades	Suture needle

Risk Factors

- **Emergency surgery; unexpected change of procedure; obesity; multiple teams; open body cavity; unusually many instruments; incorrect count; long operation; attending absent at closure.**
- **Trainee presence is associated with lower RSI risk.**
- **Human factors — leadership and communication — dominate the contributing causes** in sentinel-event reviews.

Prevention Techniques

- **Multipronged, because no strategy is foolproof.**
- **Foundation: the standardised count** of sponges and needles before, during and at close; in the US the Association of periOperative Registered Nurses (AORN) sets the standard.
- **Count discrepancy → ~100-fold odds of RSI**; if found before closure, **delay closure until the item is found.**
- **Radiographic screening** — routine or only for count discrepancy. With routine post-procedure films after abdominal-cavity cases, **most RSIs detected had correct counts** — a correct count does not exclude an RSI.

Technologic Innovations

Type	Mechanism	Performance
Counting devices	Barcoded sponges scanned in and out	Detect more counting errors; do not shorten discrepancy resolution (imaging still needed)
Detection devices	Radiofrequency tags/microchips in soft goods; handheld wand before closure	High accuracy and reliability, even in obesity

- **Choice depends on** resources, patient population, acuity and case mix.

Concept — *Counting is necessary but not sufficient: a wrong count flags danger strongly, but a right count does not guarantee safety, so counting is backed by imaging or tagging where risk is high.*

SHARPS SAFETY

- **The HIV epidemic of the 1980s** focused attention; from the late 1980s "**Universal Precautions**" (blood and body fluids from all patients) became mandated by bodies such as the Occupational Safety and Health Administration.
- **Mucous membrane or skin contact with patient blood occurs in up to half of operations.**
- **Surgeons and first assistants** sustain most injuries (up to ~60%); most are self-inflicted, but up to a quarter are inflicted by a team member.
- **About one-third of injuring devices recontact the patient** → risk of surgeon-to-patient transmission.

COMPARISON: SHARPS-INJURY COUNTERMEASURES

Measure	Rationale / evidence	Note
Double gloving	Single glove: blood contact for a large share of operating time; single-layer puncture in up to 61% of cases, on average ~40 min in ; both layers rarely breached	If both punctured, blood on a solid needle ↓ 95% — lower viral load
Blunt suture needles	For fascia and muscle closure — significantly fewer needlestick injuries	Technically satisfactory from abdominal wall closure to colonic anastomosis
Designated sharps zone	Agreed location where sharps rest → no hand-to-hand passing ; American College of Surgeons (ACS) best practice	Cut sharps injuries, blood exposure and glove tears in higher-blood-loss cases; combine with team communication; include in the time-out

Concept — *Sharps protection works at three points: a barrier (double gloves), a safer object (blunt needles) and a safer process (neutral zone) — and because a third of injuring sharps return to the patient, it protects both parties.*

FIRE SAFETY

- **Surgical fire:** on or near the patient during surgery; **~600 a year in the US.**
- **85% occur in the head, neck or upper chest; 81% under monitored anaesthesia care (MAC);** injuries disfigure, and claims involve the whole team.
- **Fire triangle** (Fig. 12.3): oxidiser + fuel + ignition in proximity; each side has an owner.

FIG. 12.3 THE "FIRE TRIANGLE" (ASSOCIATION OF PERIOPERATIVE REGISTERED NURSES)

Element	Owner	Content
Oxidiser	Anaesthesia	Not combustible itself; accelerates burning. Oxygen (95% of fires in the American Society of Anesthesiologists Closed Claims Database) and nitrous oxide — oxidising potential equal to oxygen , often directed by urologists in ambulatory procedures
Fuel	Nursing	Alcohol preps (e.g. Chloraprep, ~70% isopropyl alcohol) — dry ≥3 minutes , yet ex vivo fires still occur after 3 min, more with pooling ; alcohol flames are near-invisible. Povidone-iodine and aqueous chlorhexidine are not flammable. Drapes and dressings (cotton, plastic, paper) and body hair trap oxygen
Ignition	Surgeon	Monopolar "Bovie" in ~90% ; bipolar lowers but does not abolish risk; fibre-optic light cables (cystoscopes); lasers low-risk under irrigation, but cracked fibres can ignite even in endourology

Surgical Fire Risk Assessment

- **Silverstein Fire Risk Assessment Tool — 1 point each:** (1) open oxygen source (mask, nasal cannula); (2) ignition source (diathermy, laser); (3) **site above the xiphoid. Score 1 low, 2 intermediate, 3 high.**
- **Most urology is low or intermediate** (open oxygen above the xiphoid is rare) → standard protocols.
- **Near the airway of an intubated patient** (e.g. **buccal mucosa harvest**): **FiO₂ ≤30%, stop nitrous oxide**, and give anaesthesia **3–5 minutes' warning** before diathermy.
- **Include fire risk in the time-out;** pre-agree roles; each member reassesses the element they control.

Fire Management

Warning (smoke, unexpected flash or pop) → pause and look → water/saline (wet towels for small fires) → remove drapes and burning material to the floor (hidden smouldering) → extinguisher if not out → RACE if the fire persists

- **RACE:** Rescue/remove the patient from the fire; Alert/activate alarm and call emergency services; Confine — medical gases off, doors closed; Extinguish/evacuate — use extinguisher while leaving.

Concept — *Fire needs all three sides of the triangle, and each side belongs to a different team member — so prevention is a team task, and removing any one side (low oxygen, dry prep, no diathermy) is enough.*

ENVIRONMENTAL DISTRACTIONS

- **Sources:** equipment alarms and noise, music, pagers and smartphones, corridor noise, case-irrelevant "small talk"; distracting events are frequent (many per operative hour).
- **Effects:** in aviation, impaired prospective memory and safety checks; in theatre, **technical errors, longer operations, teamwork communication errors** → restrict to the strictly necessary.
- **"Sterile cockpit"** (aviation): no non-essential talk or activity in **pre-defined critical phases** — music off, minimal traffic and shift changes, no small talk; define phases at the briefing/time-out (e.g. **hilar clamping and unclamping in partial nephrectomy**).

- **Smartphones** blur personal and professional use and are linked to poor behaviour and outcomes; **ACS guidance**: silence ringtones, avoid personal calls, redirect to the OR desk or hard-wired phone, distinct ringtones for urgent calls. A ban is unrealistic.
- **Alarms and music**: lower noise → fewer complications. Frequent non-actionable alarms cause sensory overload and **alarm fatigue** (real events missed) → **individualise alarm settings**; develop "smart alarms".
- **Music**: very common and mostly viewed positively; effect on performance is controversial and genre effects do not generalise → consider volume, genre and lyrics for the whole team.

COMMUNICATION AND TEAM-BASED FACTORS

- **Communication failure underlies up to 70% of OR adverse events**; no protocol works without it.
- **Non-technical skills are as important as technical skills**, and dedicated training improves teamwork and safety.

Basics of Communication

- **Communication = information exchange and "constructing shared realities"**.
- **Components** (Fig. 12.4): **sender, receiver, message, channel**; a message is symbolic (words, body language, gestures, visuals); channels (e-mail, face to face) differ in speed.
Sender encodes idea into symbols/medium → channel → receiver decodes meaning — each within their own context, surrounded by noise
- **Encoding and decoding depend on personal knowledge, experience and bias; overlapping contexts make communication efficient**, mismatches cause error.
- **Noise** = anything disrupting accurate encoding/decoding: physical (music), physiological (fatigue), psychological (mood).

Errors in Communication

- **Communication fails in about 30% of OR exchanges**, a third of which create safety risk (cognitive load, interruptions, tension).
- **Types**: poor timing/situational awareness; missing or inaccurate information; no purpose or resolution; wrong audience.
- **Malpractice claims**: a substantial share of harmful breakdowns are intraoperative, mostly **verbal between two individuals**, and many occur at **handoffs**.
- **Risk factors**: high cognitive load, implicit bias/assumption, **hierarchy**, diffusion of responsibility, transitions of care.

Nontechnical Skills and Teamwork Training

- **Communication, situational awareness, decision making, leadership and teamwork predict surgical success more than technical skill**, yet were not traditionally taught.
- **Situational awareness** = the "big picture": **Where have we come from? Where are we now? Where are we going?**
- **Decision making**: assess, share knowledge, risk-stratify options and contingencies, review outcomes. **Leadership**: listen, share information, guide to shared goals. **Teamwork**: a group achieving a common goal.
- **TeamSTEPPS (Team Strategies and Tools to Enhance Performance and Patient Safety)**: national standard released **2006** by the Agency for Healthcare Research and Quality (AHRQ) with the Department of Defense — evidence-based, ready-made curricula, certified trainers.

TeamSTEPPS Overview

- **Five principles** (Fig. 12.5): **team structure, communication, leadership, situation monitoring, mutual support** — interacting dynamically around the patient care team, yielding knowledge, attitudes and performance.

FIG. 12.5 TEAMSTEPPS PRINCIPLES AND TOOLS (AGENCY FOR HEALTHCARE RESEARCH AND QUALITY)

Principle	Tools
Team structure	Correct assembly; clearly defined roles and responsibilities
Communication	SBAR — situation, background, assessment, recommendation/request (urgent information); call-outs (to many at once in emergencies); check-backs (closed loop)
Leadership	Briefs (before: roles, expectations, contingencies); huddles (ad hoc regroup, situational awareness); debriefs (lessons learned)
Situation monitoring	STEP — status of patient, team members, environment, progress to goal; I'M SAFE — illness, medication, stress, alcohol/drugs, fatigue, eating and elimination (self-check)
Mutual support	Task sharing, help sought and offered without judgement; timely, respectful, specific feedback; two-challenge rule (voice a safety concern at least twice); CUS — "I am Concerned, I am Uncomfortable, this is a Safety issue" — stops the line

- **Trap:** STEP is monitoring the **situation**, I'M SAFE monitoring **oneself**; the two-challenge rule and CUS belong to **mutual support**, not communication.
- **In urology**, TeamSTEPPS training across endoscopic, laparoscopic/robotic and open cases shortened case time, improved on-time first starts and reduced safety issues; other studies show better perceived teamwork.
- **Non-technical skills must be taught and practised.** Didactic teaching alone does not change outcomes; **simulation-based team training (SBTT)** in high-fidelity environments allows mistakes without harm, with prompt feedback.
- **In situ simulation with TeamSTEPPS reduced perinatal morbidity and mortality** in crash caesarean sections; **didactic TeamSTEPPS alone equalled no training.**
- **Urological simulation:** non-technical skills taught in high-fidelity ureteroscopy simulation outperformed didactic teaching (randomised trial); a laparoscopic partial nephrectomy crisis scenario (anaphylaxis, mislabelled specimen) was rated useful, and **few participants caught the pathology error** — reinforcing checklists and closed-loop communication.
- **All levels benefit;** mixed junior/senior cohorts favour juniors; **training technical and non-technical skills together dilutes both;** benefit depends on facilitator skill and debrief quality.

Concept — *Most safety protocols fail at the point of human interaction, not design; communication is therefore trained like a technical skill — structured tools (SBAR, check-backs, CUS) replace improvisation, and immersive simulation, not lectures, changes behaviour.*

KEY POINTS

- **Time-outs and checklists** minimise human error.
- **Universal Protocol:** preprocedure verification, site marking, time-out.
- **Count discrepancy** → 100-fold risk of an RSI.
- **Double gloving, blunt needles and a designated sharps zone** reduce sharps injury.
- **Fire risk:** oxidiser, fuel and ignition source.
- **Communication errors** cause most OR adverse events.

- **Non-technical skills training** improves efficiency and reduces communication errors; **simulation** works at all levels.